

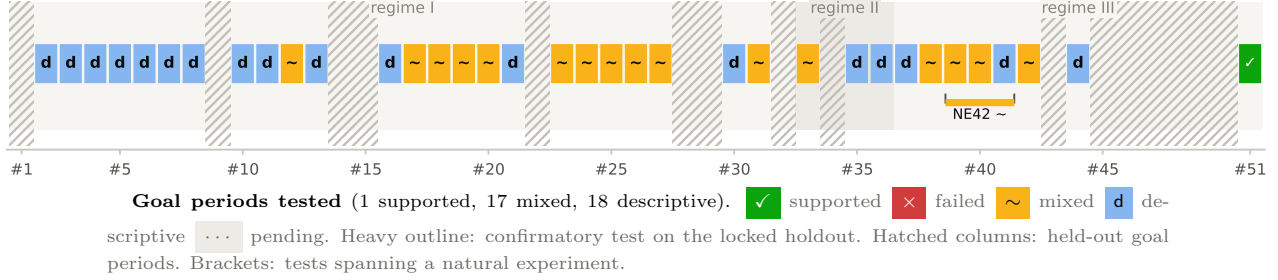
H101: Pairwise Ising vs full multi-information in co-usage

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 1.5/5

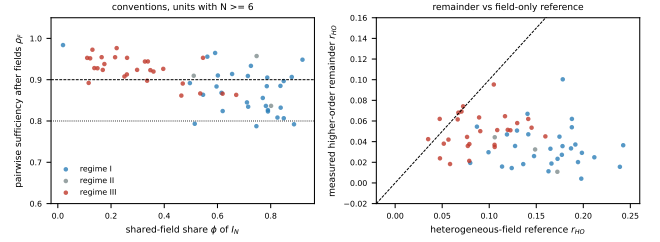
Bottom line. Convention co-usage is pairs plus fields (field-removed pairwise sufficiency 0.91, remainder 3–5% of I_N and field-sized); no separable group term is resolved, and the raw Schneidman ratio is uninformative at village sampling.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH322 asked whether co-usage of conventions has structure beyond pairs: a “group mind” would leave multi-information that a pairwise maximum-entropy model cannot capture after fields are removed. The test covers 71 non-holdout units (282 days): per day, which active agents used which H34 marker, on random 6-agent subsets. The raw Schneidman ratio is $I_2/I_N = 0.95$, but the synthetic shows this is near 1 even with a planted group term. After removing agent rates and a shared item-popularity field, pairs capture a median $\rho_F = 0.91$ of the remaining multi-information. A higher-order remainder of 3–5% of I_N is significant in 83% of units with $N \geq 6$, and it stays inside what agent-specific responses to a shared field produce in 92%. Known teams and rooms appear as signed pairwise couplings. The holdout is not run.



Observables. Units with $N \geq 6$. Left: pairwise sufficiency after field removal against the field share. Regime-I units (blue) carry most multi-information in the uniform field ($\phi \approx 0.7$); regime-III units (red) carry most in pairs ($\phi \approx 0.26$) and sit at $\rho_F \approx 0.93$. Right: the measured higher-order remainder against the remainder of a heterogeneous-field model fitted to the same subsets; almost every unit lies below the diagonal.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Models 01 and 12. Sample: one marker used in agent chat on one day; spin $s_i = 1$ if agent i used it. Fields h_i (agent-day rate), $V(K)$ (shared item popularity, $K = \text{users}$), couplings J_{ij} . Models are fitted exactly on the $2^6 - 1$ patterns with $K \geq 1$; entropies are bias-corrected by a parametric bootstrap from the fitted K-pairwise model.

$$P(\mathbf{s}) \propto \exp\left(\sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j + V(K)\right), \quad K = \sum_i s_i \geq 1$$

$$I_N = S_1 - S_N, \quad \rho_{\text{raw}} = \frac{S_1 - S_2}{S_1 - S_N}, \quad \phi = \frac{S_1 - S_K}{S_1 - S_N}$$

$$\rho_F = \frac{S_K - S_{2K}}{S_K - S_N}, \quad r_{HO} = \frac{S_{2K} - S_N}{S_1 - S_N}$$

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For a monitor: co-usage of names, numbers and coinages across an agent swarm is described by who-pairs-with-whom plus shared hot topics. Rooms and teams show up as pairwise blocks (opponents avoid each other’s terms: $J = -0.60$ vs $+0.15$ within teams). Nothing here needs a group-level interaction, so a pairwise model of co-usage is a sufficient compressed description for steering or auditing. The cynical reading is that this instrument cannot see a group mind of moderate size: a heterogeneous shared field leaves the same remainder. Raw $I_2/I_N \approx 1$ is not evidence of anything at these rates.

RESULTS VS PREDICTIONS

FAITHFULNESS SCORECARD

Prediction (dated)	Observed	Verdict
Y1–Y5 synthetic	bias correction recovers ρ_F 0.93–1.00; planted group term never gives $\rho_F < 0.8$; raw ratio uninformative	2 pass, 1 fail
R1 $\rho_F \geq 0.9$ in $\geq 70\%$	median 0.91; 56% of resolved units	mixed
R2 field share $\phi \geq 0.5$	0.56 overall; I 0.71, III 0.26	supported overall
R3 raw $I_2/I_N \geq 0.9$	0.95	uninformative
R4 remainder in $\leq 20\%$	83% of $N \geq 6$ units (pairwise world $\leq 33\%$)	failed
R6 ϕ larger in regime III	0.26 vs 0.71 (smaller)	failed
R7/R8 (A1) remainder common, field-sized	83%; $\leq$ field reference in 92%	supported
R5 projects	15 resolved; 5 candidates, days pooled	mixed
Natives G12, NE42, G51	teams, rooms = signed pair blocks (p 0.0005); Roudi size trend	mixed / mixed / supported

A mapping 1 F identifiability 1
 B assumptions 1 G ground truth 2
 C adequacy 1 H comparative 1
 D unfitted predictions 1 I transfer 0
 E interventional 1
 Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

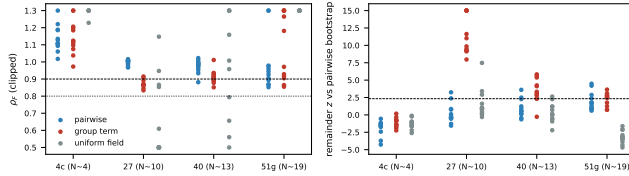
CAVEATS

A group interaction of the planted size and a heterogeneous shared field give the same remainder, so the egreore question is open, not closed. Markers within one message are not independent, so item-level resampling understates uncertainty; ratios carry no CI in the estimates table. Subsets of 6 agents marginalize hidden agents, which creates real subset-level higher order. Units with 4 agents (most of 2025) have no power. Projects pool days, so their day field is not removed.

ROUND 2

Fit a latent-factor max-ent rival (pairs plus agent-specific field loadings); resample by message; confirm C1–C4 on #22, #28, #43.

- **Essence.** whether co-usage needs any interaction beyond pairs once agent-specific responses to shared fields are modelled.
- **H101-R1** Latent-factor max-ent.
- **H101-R2** Message-level resampling.
- **H101-R3** Projects with day fields.
- **H101-R4** Debate pooling.



Second figure. Synthetic validation at the real day counts, item counts and agent rates of four units. Left: field-removed sufficiency in pairwise (blue), planted group (red) and uniform-field (grey) worlds. Right: the remainder z against the pairwise bootstrap. The group term is detected at $N \geq 10$, never at $N = 4$, and never pushes ρ_F below 0.8.

GOAL PERIODS IN DETAIL

G02 d descriptive
 G03 d descriptive
 G04 d descriptive
 G05 d descriptive
 G06 d descriptive
 G07 d descriptive
 G08 d descriptive
 G10 d descriptive
 G11 d descriptive
 G12 ~ mixed

+26 more periods: see the card.

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